THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS CHAPTER TEST SET A

Chapter 7: **A Tale of Three Intersecting Lines** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Two angles of a triangle are 50° and 60°. The third angle is: 1

(a) 110° **(b)** 40° **(c)** 70° **(d)** 130°

2. Two angles of a triangle are 64° and 38°. The third angle is: 1

(a) 26° **(b)** 78° **(c)** 102° **(d)** 116°

3. An exterior angle of a triangle is 105°. If one interior opposite angle is 45°, the other is: 1

(a) 60° **(b)** 75° **(c)** 120° **(d)** 150°

4. Sides 3 cm, 4 cm and 5 cm form a triangle? 1

(a) No **(b)** Cannot say **(c)** Only if equal **(d)** Yes

5. Sides 4 cm, 4 cm and 9 cm form a triangle? 1

(a) Only if equal **(b)** No **(c)** Yes **(d)** Cannot say

6. In an isosceles triangle the vertex angle is 70°. Each base angle is: 1

(a) 70° **(b)** 55° **(c)** 110° **(d)** 20°

7. A triangle with one angle 90° is: 1

(a) acute-angled **(b)** obtuse-angled **(c)** right-angled **(d)** equilateral

8. The three altitudes of a triangle meet at a point called: 1

(a) incentre **(b)** orthocentre **(c)** circumcentre **(d)** centroid

9. The longest side of a right triangle is called the: 1

(a) base **(b)** hypotenuse **(c)** median **(d)** altitude

10. The exterior angle of a triangle is always: 1

(a) less than each

(b) equal to 180°

(c) greater than each interior opposite angle

(d) equal to 90°

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. State the angle sum property of a triangle. 2

12. Find the third angle of a triangle with angles 52° and 68°. 2

13. Define an altitude and a median. 2

14. Can a triangle have sides 4 cm, 5 cm and 10 cm? Why? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The angles of a triangle are in ratio 2 : 3 : 4. Find them. 3

16. An exterior angle of a triangle is 110° and one interior opposite angle is 45°. Find the other two angles. 3

17. In an isosceles triangle the equal angles are each 4 times the third angle. Find the angles. 3

OR

Check whether sides 5 cm, 6 cm and 12 cm form a triangle. Which side must be reduced? 3

18. Describe the steps to construct an equilateral triangle of side 5 cm. 3

OR

A triangle has angles (x + 10)°, (2x)° and (3x − 10)°. Find x and the angles. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. (i) Prove the angle sum property by tearing the corners. 5
(ii) Find the missing angle if two are 63° and 47°.
(iii) Name the triangle.

OR

Case study: A roof truss is an isosceles triangle with apex 100°. 5
(i) Find base angles.
(ii) Find the exterior angle at the apex.
(iii) Name the type of triangle.
(iv) The two equal rafters are 4 m each; find the perimeter if the base is 6 m.

20. (i) Construct a triangle with sides 4 cm, 5 cm, 6 cm — list the steps. 5
(ii) Draw an altitude from the 6 cm side.
(iii) Mark the orthocentre.

OR

(i) State the triangle inequality. 5
(ii) Which sets form triangles: (3,4,8), (5,6,10), (2,2,3)?
(iii) Justify one.

END OF QUESTION PAPER — Total: 40 Marks

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